

DCCN — 10-Page Master Quick Revision

High-Yield Formula Sheet, Protocol Matrices & Top 10 BIT Mesra PYQ Solutions

⚡ 10-Page Master Quick Revision — Data Communication & Computer Networks (CS24305)

Page 1: Complete 52-Topic Syllabus Progress Checklist (Modules I – V)

Page 3: Physical Layer: Nyquist Rate, Shannon Capacity & Waveform Rules

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Page 10: Application Services: DNS, HTTP, DHCP DORA & URL Life Cycle

The Complete 52–Topic Syllabus Master Checklist

52 / 52 Syllabus Topics Complete

Module	Topic #	Topic Name	Status
M1: Overview (13)	1	Communication Model (S-M-M-P)	✓ Complete
	2	Data Communications (Simplex, Half, Full-Duplex) [UPLOADED PYQ]	✓ Complete
	3	Networks (PAN, LAN, MAN, WAN)	✓ Complete
	4	Internet Architecture & Packet Switching	✓ Complete
	5	OSI 7-Layer Reference Model & Layer Functions	✓ Complete
	6	TCP/IP Protocol Architecture (OSI vs. TCP/IP) [UPLOADED PYQ]	✓ Complete
	7	Standards, Protocol Layers & Encapsulation	✓ Complete
	8	Internet Applications (HTTP, SMTP, DNS, DHCP)	✓ Complete
	9	Data Transmission Concepts (Bit vs. Baud, Latency)	✓ Complete
	10	Analog Data Transmission (Amplitude, Frequency, Phase)	✓ Complete
	11	Digital Data Transmission (Discrete Levels vs. Noise)	✓ Complete
	12	Transmission Impairments (Attenuation, Distortion, Noise) [UPLOADED PYQ]	✓ Complete
	13	Channel Capacity (Nyquist & Shannon Theorems) [UPLOADED PYQ]	✓ Complete
M2: Media & Signals (6)	14	Guided Transmission Media (Twisted Pair, Coaxial, Optical Fiber)	✓ Complete
	15	Wireless Media & Propagation (Ground, Sky, LOS, Satellite) [UPLOADED PYQ]	✓ Complete
	16	Digital Signaling (Line Codes: NRZ, Manchester, AMI) [UPLOADED PYQ]	✓ Complete
	17	Analog Signaling (Carrier Wave Parameters)	✓ Complete
	18	Encoding Techniques (Waveform Comparison Matrix)	✓ Complete
	19	Modulation Techniques (ASK, FSK, PSK, QAM)	✓ Complete
M3: Errors & Link (8)	20	Types of Errors (Single-Bit vs. Burst Errors)	✓ Complete

Module	Topic #	Topic Name	Status
	21	Error Detection Techniques (Parity & Checksum) [UPLOADED PYQ]	✓ Complete
	22	Checksum Algorithms (1's Complement Summation)	✓ Complete
	23	Cyclic Redundancy Check (CRC-32 Modulo-2 Division) [UPLOADED PYQ]	✓ Complete
	24	Error Correction (Hamming Code $2^r \geq m+r+1$ & Syndrome)	✓ Complete
	25	Flow Control (Stop-and-Wait vs. Sliding Window)	✓ Complete
	26	Error Control ARQs (Stop-and-Wait, GBN, Selective Repeat) [UPLOADED PYQ]	✓ Complete
	27	HDLC Protocol (Flag 01111110, I/S/U-Frames)	✓ Complete
M4: WAN & LAN (9)	30	Switching Networks (Circuit vs. Packet Switching) [UPLOADED PYQ]	✓ Complete
	31	Circuit-Switching Networks (Setup, Transfer, Teardown)	✓ Complete
	32	Circuit-Switching Concepts (Blocking & Dedicated Paths)	✓ Complete
	33	Packet-Switching Principles (Datagram vs. Virtual Circuit) [UPLOADED PYQ]	✓ Complete
	34	Cellular Network Principles (Frequency Reuse, Handoff)	✓ Complete
	35	Cellular Generations Evolution (1G Voice to 5G Ultra-Low Latency)	✓ Complete
	36	Network Topologies (Bus, Star, Ring, Mesh, Tree)	✓ Complete
	37	LAN Protocol Architecture (IEEE 802: LLC vs. MAC)	✓ Complete
	38	Virtual Local Area Networks (VLAN 802.1Q)	✓ Complete
M5: IP, Routing & Apps (16)	39	Traditional Ethernet (IEEE 802.3 CSMA/CD at 10 Mbps)	✓ Complete
	40	High-Speed Ethernet (Fast 100M, Gigabit 1G, 10G)	✓ Complete
	41	IEEE 802.11 Wi-Fi (CSMA/CA with RTS/CTS Handshake)	✓ Complete
	42	Internet Protocol (IPv4 20-Byte Header vs. IPv6 128-Bit) [UPLOADED PYQ]	✓ Complete

Module	Topic #	Topic Name	Status
	43	IP Addressing & Subnetting (2^{borrowed} & $2^{\text{host}} - 2$)	✓ Complete
	44	Transport Protocols (TCP 3-Way Handshake vs. UDP) [UPLOADED PYQ]	✓ Complete
	45	Routing in Packet Networks (Hop Count, Metric Costs)	✓ Complete
	46	Distance Vector Routing (Bellman-Ford & RIP Count-to-Infinity)	✓ Complete
	47	Link State Routing (Dijkstra's Algorithm & OSPF)	✓ Complete
	48	Path Vector Routing (Border Gateway Protocol BGP-4)	✓ Complete
	49	Congestion Control (Leaky Bucket vs. Token Bucket)	✓ Complete
	50	Traffic Management & Quality of Service (QoS Dimensions)	✓ Complete
	51	Simple Mail Transfer Protocol (SMTP, POP3, IMAP)	✓ Complete
	52	Domain Name System (DNS Hierarchy & Records A, MX)	✓ Complete
	53	Hypertext Transfer Protocol (HTTP/1.1 vs. HTTP/2 & Statuses)	✓ Complete
	54	Dynamic Host Configuration Protocol (DHCP DORA Sequence)	✓ Complete

★ Ultra-High-Yield Top 20 DCCN Exam Topics

#	Topic	Core Concept / Formula
1	OSI 7-Layer Model	All People Seem To Need Data Processing
2	TCP/IP vs. OSI	4-layer pragmatic architecture vs 7-layer theoretical reference
3	Guided Media	Twisted Pair (UTP/STP), Coaxial Cable, Optical Fiber (Total Internal Reflection)
4	Manchester Encoding	Mandatory mid-bit clock transition (Self-clocking)
5	Nyquist & Shannon Capacity	$C = 2B \log_2 L$ (Noiseless); $C = B \log_2(1 + \text{SNR})$ (Noisy)
6	CRC-32 Modulo-2 Division	Append r zeros, divide by generator $G(x)$, remainder is CRC
7	Hamming Code	$2^r \geq m + r + 1$, parity check positions at powers of 2
8	Stop-and-Wait vs GBN vs SR	$W_s = 1$ vs $W_s \leq 2^m - 1$ vs $W_s \leq 2^{m-1}$
9	HDLC Frame Formats	Flag `01111110`, Information (I), Supervisory (S), Unnumbered (U)
10	FDM vs. TDM	Frequency bands with guard bands vs. time slots (Synchronous vs Statistical)
11	Circuit vs. Packet Switching	Dedicated 3-phase path vs. dynamic statistical packet multiplexing
12	Cellular Networks (1G-5G)	Frequency reuse $D = R\sqrt{3N}$, Handoff, 1G Analog → 5G Low-Latency
13	Network Topologies	Mesh ($\frac{N(N-1)}{2}$ links), Star, Bus, Ring, Tree
14	VLAN (IEEE 802.1Q)	Logical segmentation, broadcast domain reduction
15	IPv4 Header Anatomy	20-byte base header (Version, IHL, Total Length, TTL, Protocol, Checksum)
16	Subnetting Calculations	Subnets = 2^{borrowed} , Usable Hosts = $2^{\text{host}} - 2$
17	TCP vs. UDP	Reliable connection-oriented byte stream vs Unreliable best-effort datagram
18	Routing: RIP vs OSPF vs BGP	Bellman-Ford (DV) vs Dijkstra (Link-State) vs Path Vector (BGP)
19	Congestion: Leaky vs Token	Strict constant output rate vs Regulated burst capacity
20	URL → Webpage Journey	DHCP (DORA) → DNS → ARP → TCP 3-Way → TLS → HTTP → IP Routing